

GOOD MORNING!

早上好!

안녕하세요!

DAY 7



DAY 6 (FINAL PROJECT)

- 시스템 설계에 기반한 객체 감지 모델 구현
 - 로봇 환경에 적용 및 Unit Test
 - 모듈로 제작하고 launch파일로 구현
 - code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑
- 시스템 설계에 기반한 SysMon 설계 구현
 - 로봇 환경에 적용 및 Unit Test
 - 모듈로 제작하고 launch파일로 구현
 - code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 7 (FINAL PROJECT)

- 시스템 설계에 기반한 **AMR 제어** 구현
- 로봇 환경에 적용 및 Unit Test
- 모듈로 제작하고 launch 파일로 구현
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 8-9 (FINAL PROJECT)

- 개별 기능 통합 구현 및 Integration 테스트
- 통합 Launch 파일로 구현
- Robust한 시스템 구축을 위한 예외 처리 및 Code Refactoring
- code 정리 및 버전관리, 문서 작성 및 영상 촬영, 팀 내 기술 브리핑

DAY 10 (FINAL PROJECT)

- 프로젝트 발표 및 시연
- 최종 산출문 정리(소스코드, 발표 PPT, 동작 영상)
- 팀 간 기술 컨퍼런스를 통한 기술 극복 경험담, 노하우 교류(채점 대상X)

프로젝트 RULE NUMBER ONE!!!

Have Fun Fun Fun!



PROJECT SCHEDULE REVIEW



BUSINESS/SOLUTION REQUIREMENT **UPDATE** BY EACH TEAM

Using the posted notes and flipchart as needed

SYSTEM REQUIREMENT **UPDATE** BY EACH TEAM

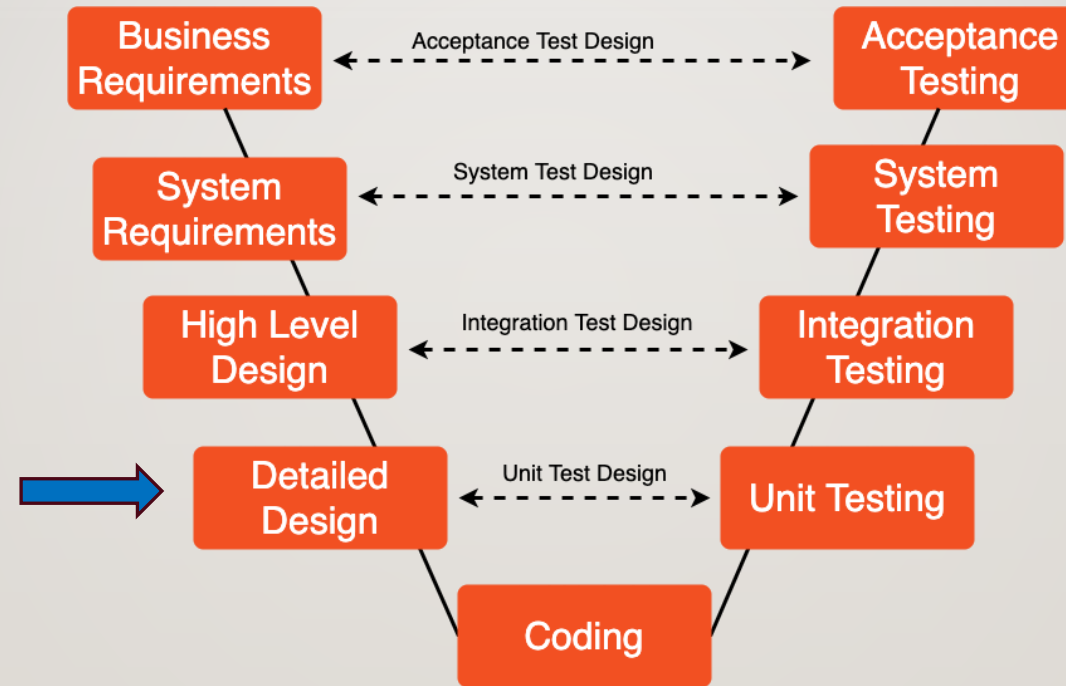
Using the posted notes and flipchart as needed

SYSTEM DESIGN ARCHITECTURE **UPDATE** BY EACH TEAM

Using the posted notes and flipchart as needed



SPRINT 3

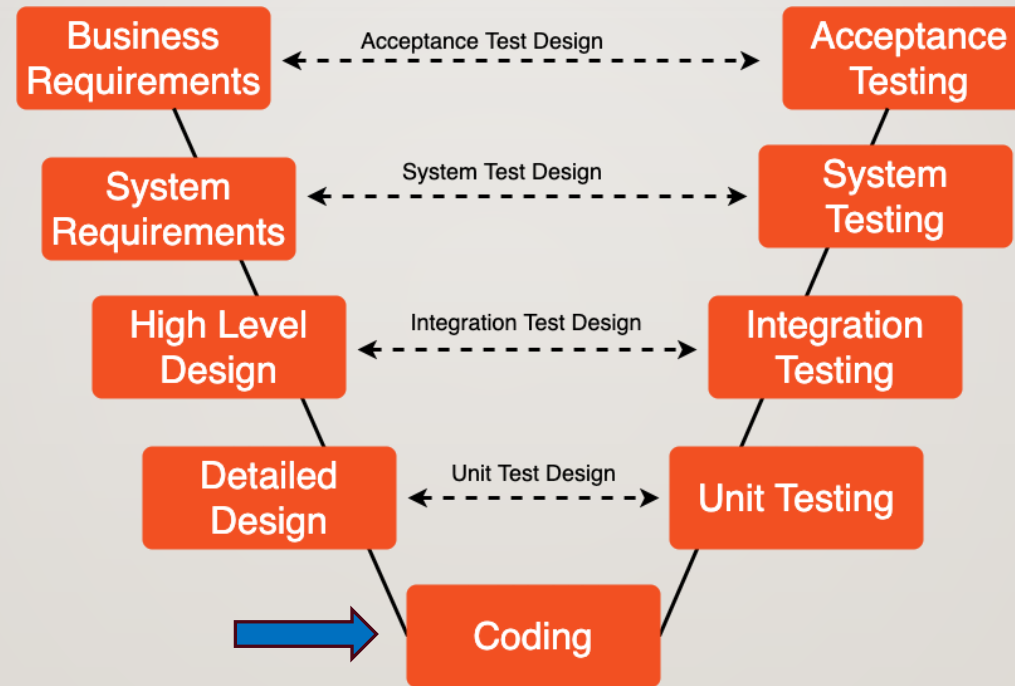


SDLC - V Model - notepub.io

TEAM EXERCISE 17

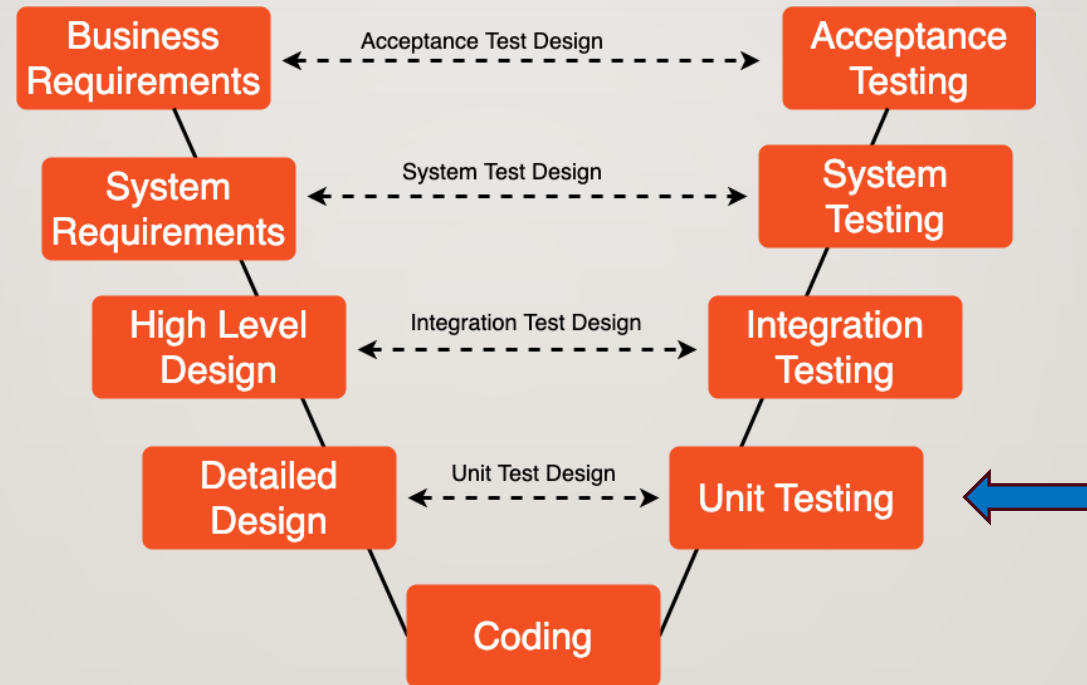
Perform Detail Design of Sprint 3 using Process Flow Diagram

SPRINT 3



SDLC - V Model - notepub.io

SPRINT 3



SDLC - V Model - notepub.io

TEAM EXERCISE 18

Perform coding and testing of Sprint 3 Module

(평가 4) DESIGN/CODE/TEST REVIEW BY EACH TEAM

Show actual results against the expected results and explain the code written

WHAT TO DO IN CODE REVIEW

- Explain
 - Design

Topic: sensor_msg:

Type: xxxx

Data_a: {int} # number of detection

Data_b: {list} # contains list of class types

...

PGM.py

Node/Class_A:

var_a = xxx

var_b = yyy

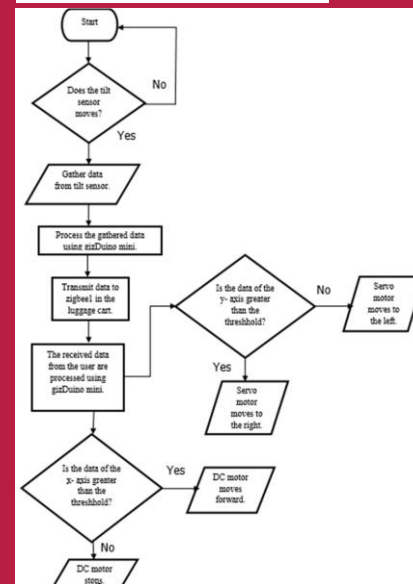
....

Main():

Call Func_A;

....

Func_A:



Topic: out_msg:

Type: xxxx

Data_a: {Bool} # True or False

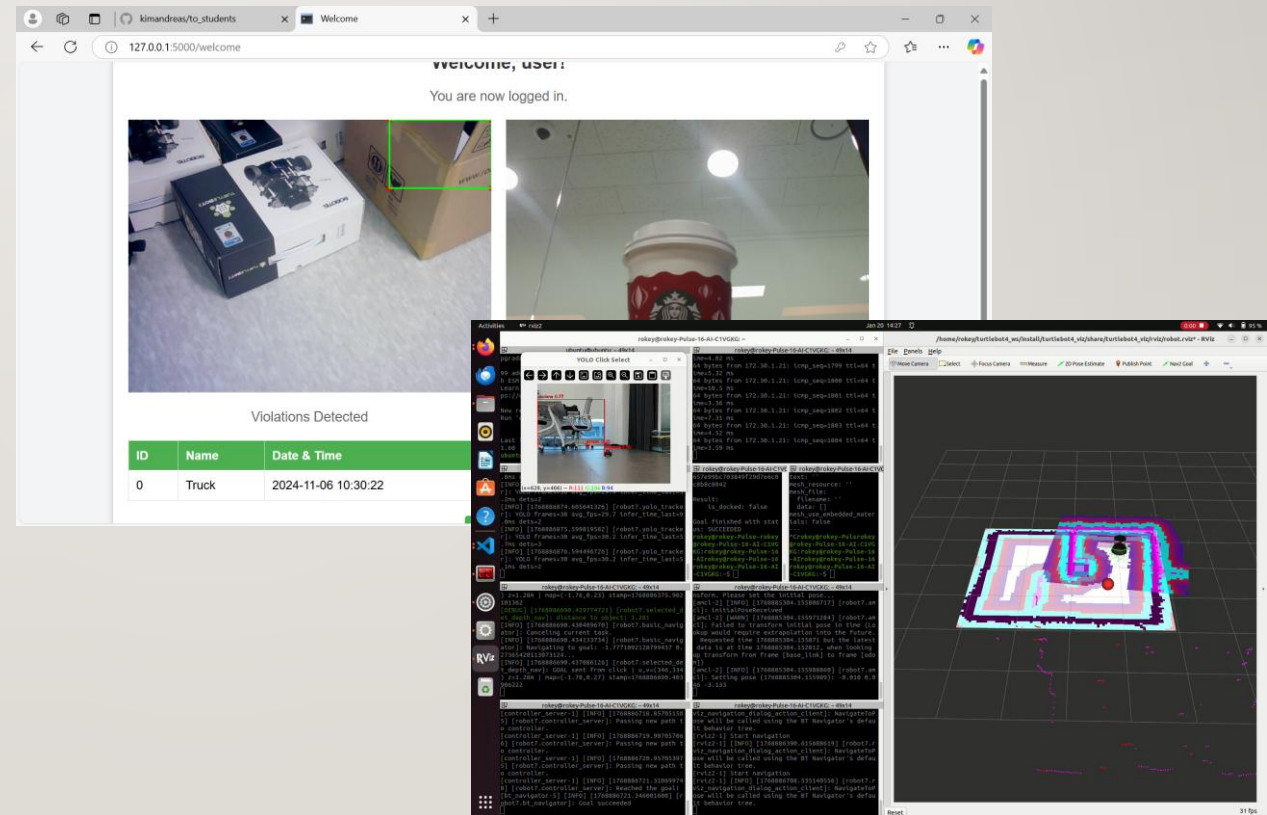
WHAT TO DO IN CODE REVIEW

- Explain
 - Design
 - Code

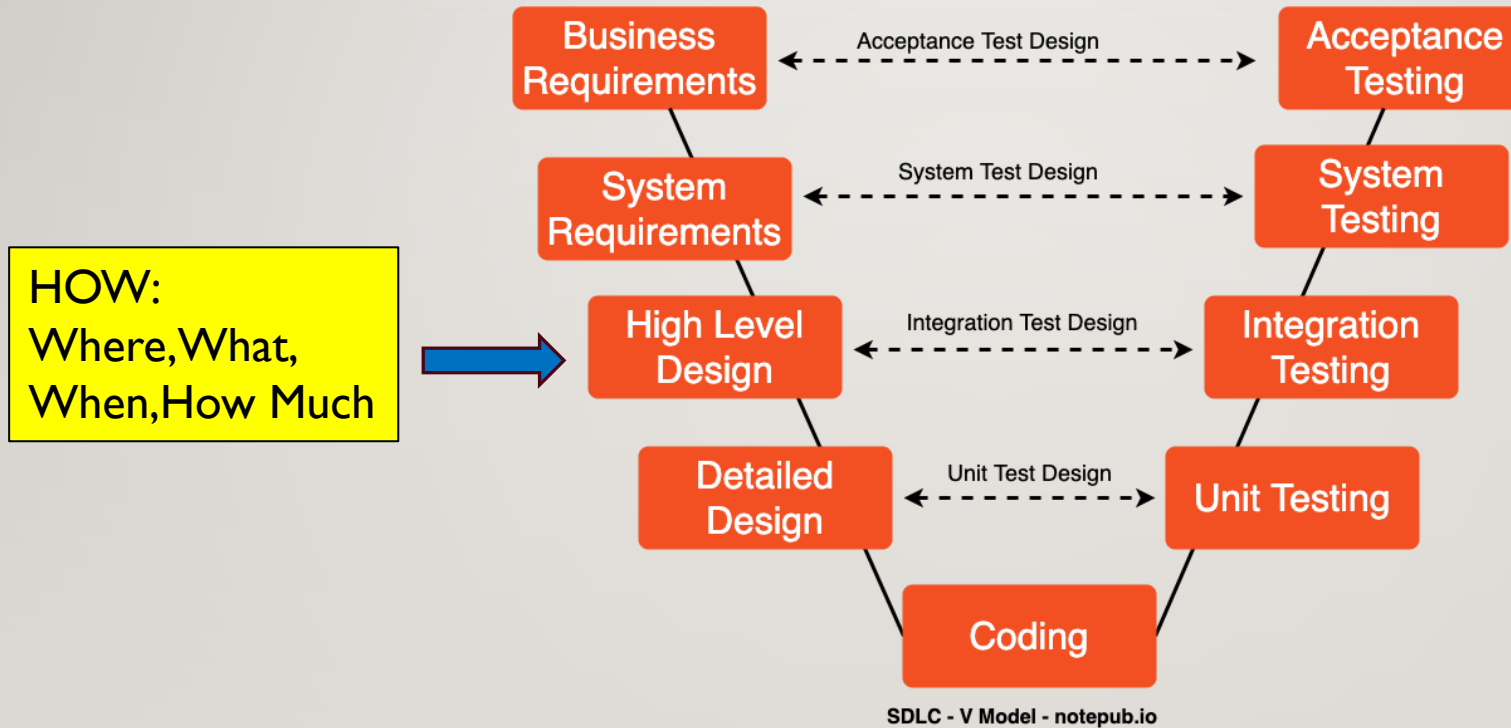
```
src > day3 > day3 > 3_1_c_follow_waypoints.py > main
12 # distributed under the
13 # WITHOUT WARRANTIES OR
14 # See the License for the specific language governing permissions and
15 # limitations under the License.
16 #
17 # @author Roni Kreinin (rkreinin@clearpathrobotics.com)
18
19 import rcipy
20
21 from turtlebot4_navigation.turtlebot4_navigator import TurtleBot4Directions, TurtleBot4Navigator
22
23
24 def main():
25     rcipy.init()
26     (variable) navigator: TurtleBot4Navigator
27     navigator = TurtleBot4Navigator()
28
29     # Start on dock
30     if not navigator.getDockedStatus():
31         navigator.info('Docking before intialising pose')
32         navigator.dock()
33
34     # Set initial pose
35     initial_pose = navigator.getPoseStamped([0.0, 0.0], TurtleBot4Direction.NORT
36     navigator.setInitialPose(initial_pose)
37
38     # Wait for Nav2
39     navigator.waitUntilNav2Active()
40
41 # Set goal poses
```


WHAT TO DO IN CODE REVIEW

- Explain
 - Design
 - Code
 - Test/Validation
 - Test Code
 - Log Output
 - Screenshot/video
 - Photo/Video
 -



SW DEVELOPMENT PROCESS

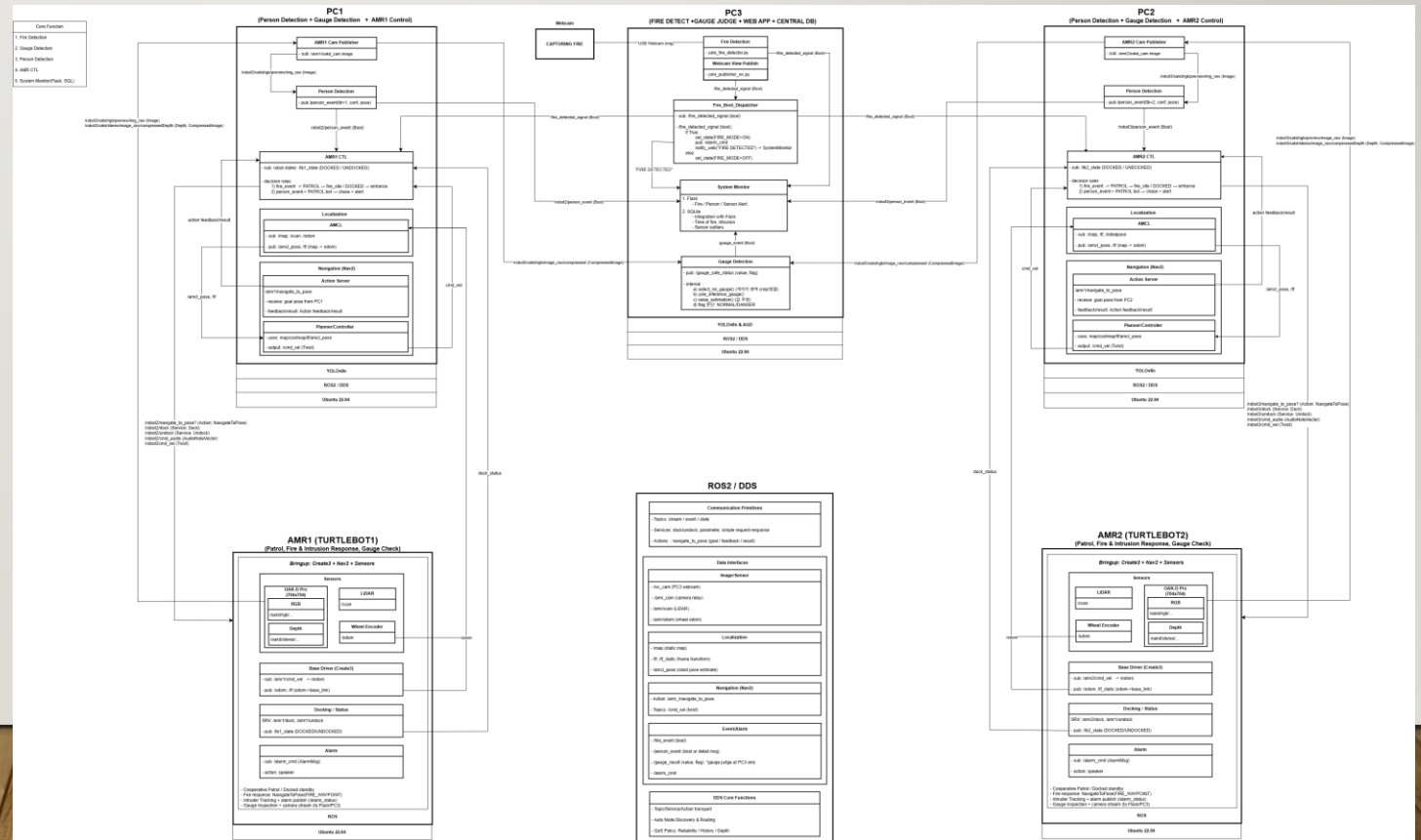


VISUALIZATION – SYSTEM FUNCTIONAL PROCESS FLOW (ARCHITECTURE) DIAGRAMS

- To-Be Functional Process Flow Diagram

Detection Alert
AMR Controller

- Functions
 - Interfaces
- Dataflow



(평가 5) SYSTEM DESIGN ARCHITECTURE DIAGRAM PRESENTATION BY EACH TEAM



SEND SYSTEM DESIGN DOC. HERE

산출물 구글드라이브

- https://drive.google.com/drive/folders/IAmPeoNG_nFoKaYlQh-Iw9OTtoDhEkHTW?usp=sharing

Before: Day 10, 9:30 a.m.

- ***Don't forget to include your Test and Integration Plan in the Doc.***

CHECK SOLUTION NETWORK SETTING

DAY 5 - System Monitor & Multi Robot

Aa 이름

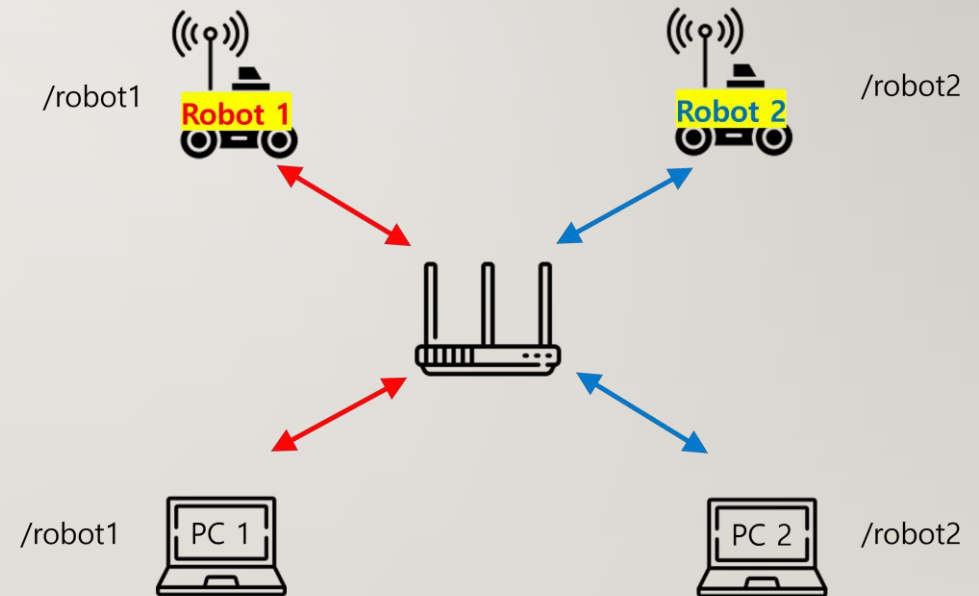
☀ 상태

● Multi Robot Standard Setup

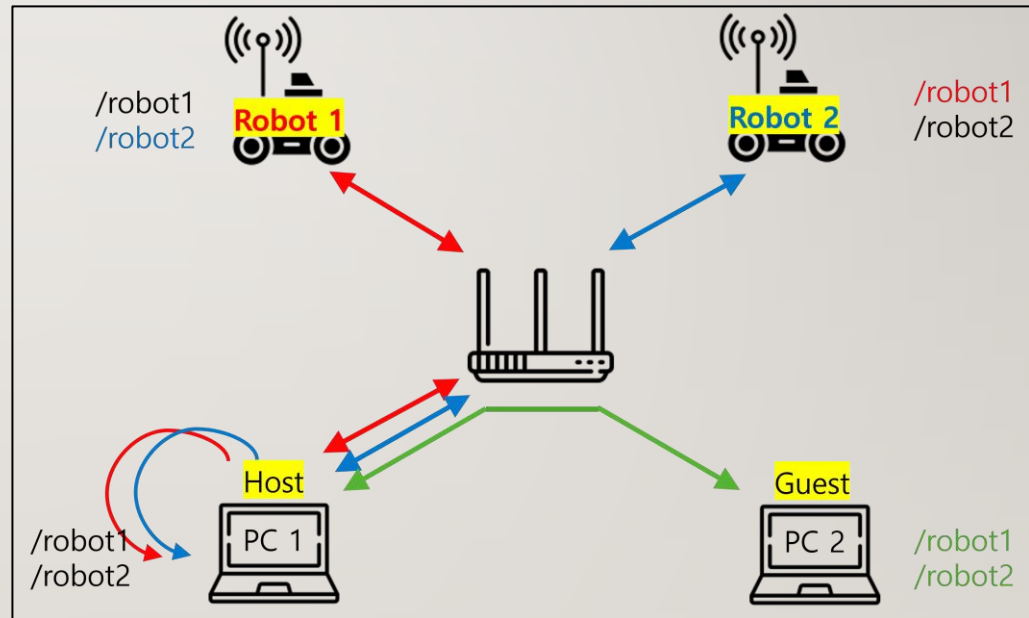
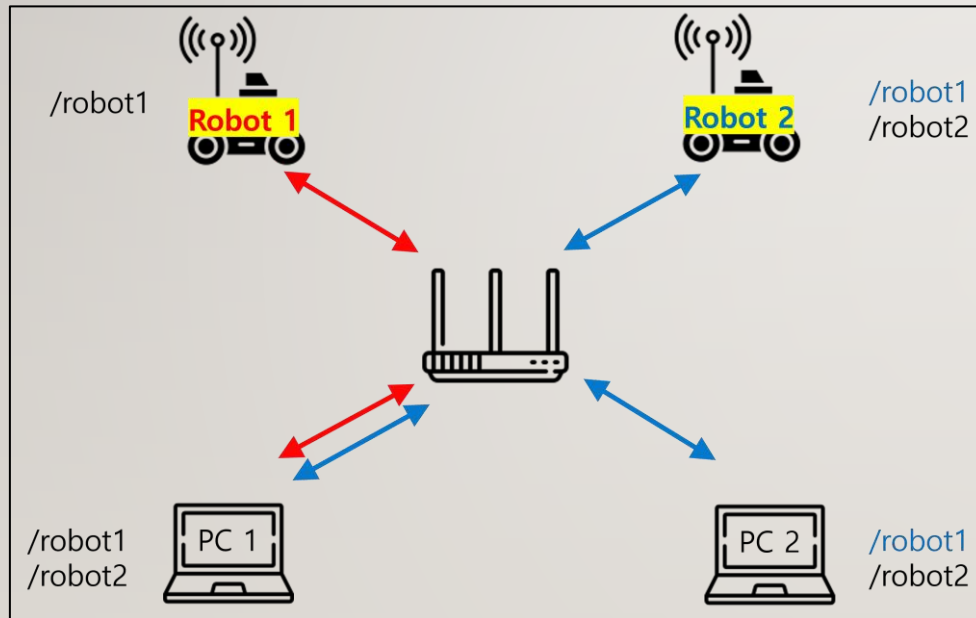
● 시작 전

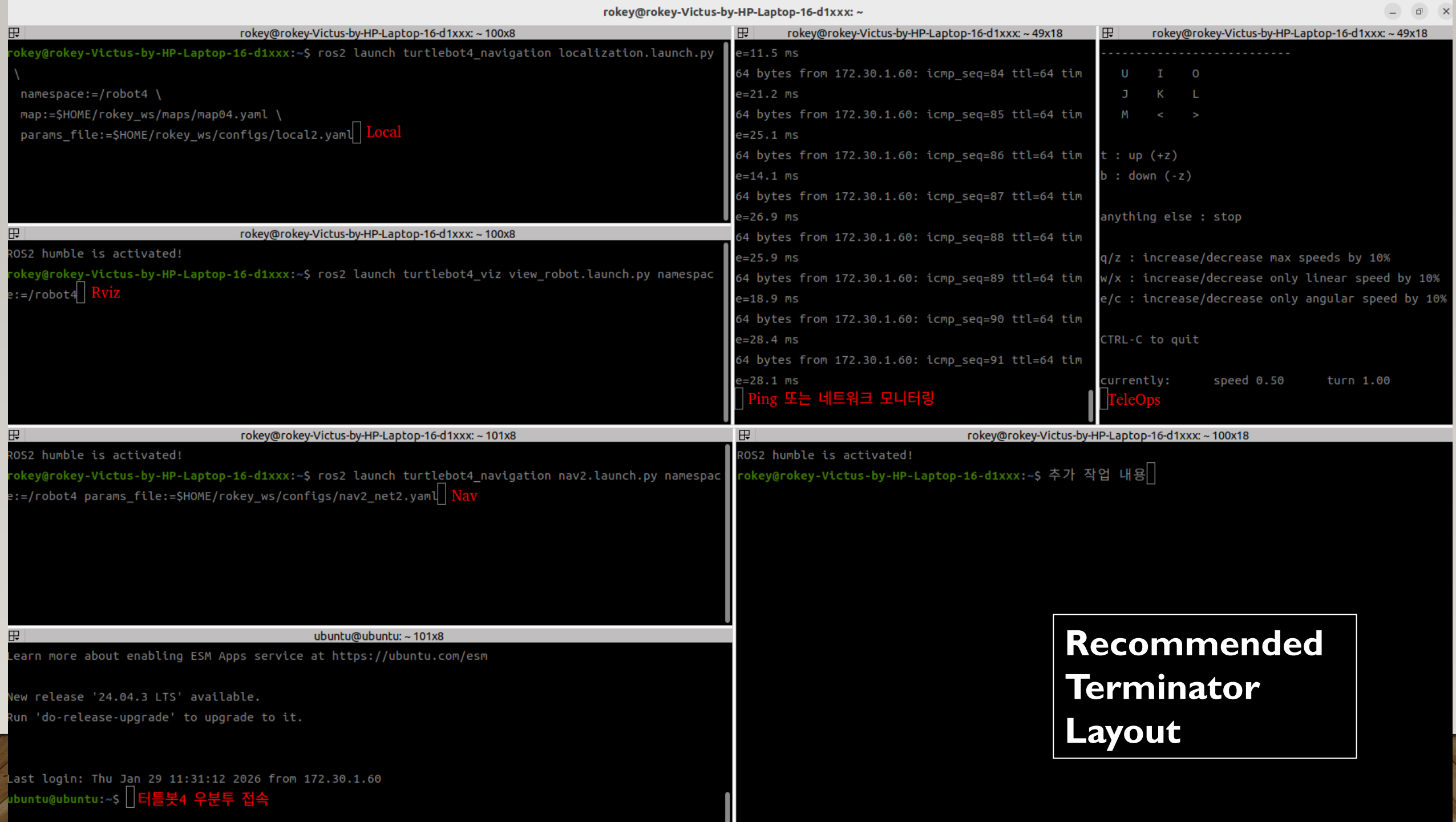
● Multi Robot Custom Discovery Setup

● 시작 전



CHECK SOLUTION NETWORK SETTING





프로젝트 RULE NUMBER ONE!!!

Are we having
Fun???

